34. It is not commutative. Let $f(x) = x^2$ and $g(x) = x + 1$. Then $(f \circ g)(3) = f(g(3)) = f(4) = 16$ but $(g \circ f)(3) = g(f(3)) = g(9) = 10$.

35. It is not true. Let $*$ be $+$ and let $*'$ be $\cdot$ and let $S = \mathbb{Z}$. Then $2 + (3 \cdot 5) = 17$ but $(2+3) \cdot (2+5) = 35$.

36. Let $a, b \in H$. By definition of H, we have $a * x = x * a$ and $b * x = x * b$ for all $x \in S$. Using the fact that $*$ is associative, we then obtain, for all $x \in S$,

$$(a * b) * x = a * (b * x) = a * (x * b) = (a * x) * b = (x * a) * b = x * (a * b).$$

This shows that $a * b$ satisfies the defining criterion for an element of H, so $(a * b) \in H$.

37. Let $a, b \in H$. By definition of H, we have $a * a = a$ and $b * b = b$. Using, one step at a time, the fact that $*$ is associative and commutative, we obtain

$$\begin{aligned}(a * b) * (a * b) &= [(a * b) * a] * b = [a * (b * a)] * b = [a * (a * b)] * b \\ &= [(a * a) * b] * b = (a * b) * b = a * (b * b) = a * b.\end{aligned}$$

This show that $a * b$ satisfies the defining criterion for an element of H, so $(a * b) \in H$.

3. Isomorphic Binary Structures

1. i) ϕ must be one to one. ii) $\phi[S]$ must be all of S'. iii) $\phi(a * b) = \phi(a) *' \phi(b)$ for all $a, b \in S$.

2. It is an isomorphism; ϕ is one to one, onto, and $\phi(n+m) = -(n+m) = (-n) + (-m) = \phi(n) + \phi(m)$ for all $m, n \in \mathbb{Z}$.

3. It is not an isomorphism; ϕ does not map $\mathbb{Z}$ onto $\mathbb{Z}$. For example, $\phi(n) \neq 1$ for all $n \in \mathbb{Z}$.

4. It is not an isomorphism because $\phi(m+n) = m+n+1$ while $\phi(m)+\phi(n) = m+1+n+1 = m+n+2$.

5. It is an isomorphism; ϕ is one to one, onto, and $\phi(a+b) = \frac{a+b}{2} = \frac{a}{2} + \frac{b}{2} = \phi(a) + \phi(b)$.

6. It is not an isomorphism because ϕ does not map $\mathbb{Q}$ onto $\mathbb{Q}$. $\phi(a) \neq -1$ for all $a \in \mathbb{Q}$.

7. It is an isomorphism because ϕ is one to one, onto, and $\phi(xy) = (xy)^3 = x^3y^3 = \phi(x)\phi(y)$.

8. It is not an isomorphism because ϕ is not one to one. All the 2×2 matrices where the entries in the second row are double the entries above them in the first row are mapped into 0 by ϕ.

9. It is an isomorphism because for 1×1 matrices, $[a][b] = [ab]$, and $\phi([a]) = a$ so ϕ just removes the brackets.

10. It is an isomorphism. For any base $a \neq 1$, the exponential function $f(x) = a^x$ maps $\mathbb{R}$ one to one onto $\mathbb{R}^+$, and ϕ is the exponential map with $a = 0.5$. We have $\phi(r+s) = 0.5^{(r+s)} = (0.5^r)(0.5^s) = \phi(r)\phi(s)$.

11. It is not an isomorphism because ϕ is not one to one; $\phi(x^2) = 2x$ and $\phi(x^2+1) = 2x$.

12. It is not an isomorphism because ϕ is not one to one: $\phi(\sin x) = \cos 0 = 1$ and $\phi(x) = 1$.

13. No, because ϕ does not map F onto F. For all $f \in F$, we see that $\phi(f)(0) = 0$ so, for example, no function is mapped by ϕ into $x + 1$.

14. It is an isomorphism. By calculus, $\phi(f) = f$, so ϕ is the identity map which is always an isomorphism of a binary structure with itself.

15. It is not an isomorphism because ϕ does not map F onto F. Note that $\phi(f)(0) = 0 \cdot f(0) = 0$. Thus there is no element of F that is mapped by ϕ into the constant function 1.

16. **a.** For ϕ to be an isomorphism, we must have

$$m * n = \phi(m-1) * \phi(n-1) = \phi((m-1) + (n-1)) = \phi(m+n-2) = m+n-1.$$

The identity element is $\phi(0) = 1$.

b. Using the fact that ϕ^{-1} must also be an isomorphism, we must have

$$m * n = \phi^{-1}(m+1) * \phi^{-1}(n+1) = \phi^{-1}((m+1) + (n+1)) = \phi^{-1}(m+n+2) = m+n+1.$$

The identity element is $\phi^{-1}(0) = -1$.

17. **a.** For ϕ to be an isomorphism, we must have

$$m * n = \phi(m-1) * \phi(n-1) = \phi((m-1) \cdot (n-1)) = \phi(mn - m - n + 1) = mn - m - n + 2.$$

The identity element is $\phi(1) = 2$.

b. Using the fact that ϕ^{-1} must also be an isomorphism, we must have

$$m * n = \phi^{-1}(m+1) * \phi^{-1}(n+1) = \phi^{-1}((m+1) \cdot (n+1)) = \phi^{-1}(mn + m + n + 1) = mn + m + n.$$

The identity element is $\phi^{-1}(1) = 0$.

18. **a.** For ϕ to be an isomorphism, we must have

$$a * b = \phi\left(\frac{a+1}{3}\right) * \phi\left(\frac{b+1}{3}\right) = \phi\left(\frac{a+1}{3} + \frac{b+1}{3}\right) = \phi\left(\frac{a+b+2}{3}\right) = a+b+1.$$

The identity element is $\phi(0) = -1$.

b. Using the fact that ϕ^{-1} must also be an isomorphism, we must have

$$a * b = \phi^{-1}(3a-1) * \phi^{-1}(3b-1) = \phi^{-1}((3a-1) + (3b-1)) = \phi^{-1}(3a+3b-2) = a+b-\frac{1}{3}.$$

The identity element is $\phi^{-1}(0) = 1/3$.

19. **a.** For ϕ to be an isomorphism, we must have

$$a * b = \phi\left(\frac{a+1}{3}\right) * \phi\left(\frac{b+1}{3}\right) = \phi\left(\frac{a+1}{3} \cdot \frac{b+1}{3}\right) = \phi\left(\frac{ab+a+b+1}{9}\right) = \frac{ab+a+b-2}{3}.$$

The identity element is $\phi(1) = 2$.

b. Using the fact that ϕ^{-1} must also be an isomorphism, we must have

$$a * b = \phi^{-1}(3a-1) \cdot \phi^{-1}(3b-1) = \phi^{-1}((3a-1) \cdot (3b-1)) = \phi^{-1}(9ab - 3a - 3b + 1) = 3ab - a - b + \frac{2}{3}.$$

The identity element is $\phi^{-1}(1) = 2/3$.

20. Computing $\phi(x * y)$ is done by first executing the binary operation $*$, and then performing the map ϕ. Computing $\phi(x) *' \phi(y)$ is done by first performing the map ϕ, and then executing the binary operation $*'$. Thus, reading in left to right order of peformance, the isomorphism property is

$$\text{(binary operation)(map)} = \text{(map)(binary operation)}$$

which has the formal appearance of commutativity.

21. The definition is incorrect. It should be stated that $\langle S, * \rangle$ and $\langle S', *' \rangle$ are binary structures, ϕ must be one to one and onto S', and the universal quantifier "for all $a, b \in S$" should appear in an appropriate place.

Let $\langle S, * \rangle$ and $\langle S', *' \rangle$ be binary structures. A map $\phi : S \to S'$ is an **isomorphism** if and only if ϕ is one to one and onto S', and $\phi(a * b) = \phi(a) *' \phi(b)$ for all $a, b \in S$.

22. It is badly worded. The "for all $s \in S$" applies to the equation and not to the "is an identity for $*$".

Let $*$ be a binary operation on a set S. An element e of S is an **identity element** for $*$ if and only if $s * e = e * s = s$ for all $s \in S$.

23. Suppose that e and $\overline{e}$ are two identity elements and, viewing each in turn as an identity element, compute $e * \overline{e}$ in two ways.

24. a. Let $*$ be a binary operation on a set S. An element e_L of S is a **left identity element** for $*$ if and only if $e_L * s = s$ for all $s \in S$.

b. Let $*$ be a binary operation on a set S. An element e_R of S is a **right identity element** for $*$ if and only if $s * e_R = s$ for all $s \in S$.

A one-sided identity element is not unique. Let $*$ be defined on S by $a * b = a$ for all $a, b \in S$. Then every $b \in S$ is a right identity. Similarly, a left identity is not unique. If in the proof of Theorem 3.13, we replace e by e_L and $\overline{e}$ by $\overline{e}_L$ everywhere, and replace the word "identity" by "left identity", the first incorrect statement would be, "However, regarding $\overline{e}_L$ as left identity element, we must have $e_L * \overline{e}_L = e_L$."

25. No, if $\langle S* \rangle$ has a left identity element e_L and a right identity element e_R, then $e_L = e_R$.

Proof Because e_L is a left identity element we have $e_L * e_R = e_R$, but viewing e_R as right identity element, $e_L * e_R = e_L$. Thus $e_L = e_R$.

26. *One-to-one:* Suppose that $\phi^{-1}(a') = \phi^{-1}(b')$ for $a', b' \in S'$. Then $a' = \phi(\phi^{-1}(a')) = \phi(\phi^{-1}(b')) = b'$, so ϕ^{-1} is one to one.

Onto: Let $a \in S$. Then $\phi^{-1}(\phi(a)) = a$, so ϕ^{-1} maps S' onto S.

Homomorphism property: Let $a', b' \in S'$. Now

$$\phi(\phi^{-1}(a' *' b')) = a' *' b'.$$

Because ϕ is an isomorphism,

$$\phi(\phi^{-1}(a') * \phi^{-1}(b\ ,')) = \phi(\phi^{-1}(a')) *' \phi(\phi^{-1}(b')) = a' *' b'$$

also. Because ϕ is one to one, we conclude that

$$\phi^{-1}(a' * b') = \phi^{-1}(a') *' \phi^{-1}(b').$$

27. *One-to-one:* Let $a, b \in S$ and suppose $(\psi \circ \phi)(a) = (\psi \circ \phi)(b)$. Then $\psi(\phi(a)) = \psi(\phi(b))$. Because ψ is one to one, we conclude that $\phi(a) = \phi(b)$. Because ϕ is one to one, we must have $a = b$.

Onto: Let $a'' \in S''$. Because ψ maps S' onto S'', there exists $a' \in S'$ such that $\psi(a') = a''$. Because ϕ maps S onto S', there exists $a \in S$ such that $\phi(a) = a'$. Then $(\psi \circ \phi)(a) = \psi(\phi(a)) = \psi(a') = a''$, so $\psi \circ \phi$ maps S onto S''.

Homomorphism property: Let $a, b \in S$. Since ϕ and ψ are isomorphisms, $(\psi \circ \phi)(a * b) = \psi(\phi(a * b)) = \psi(\phi(a) *' \phi(b)) = \psi(\phi(a)) *'' \psi(\phi(b)) = (\psi \circ \phi)(a) *'' (\psi \circ \phi)(b)$.

28. Let $\langle S, * \rangle, \langle S', *' \rangle$ and $\langle S'', *'' \rangle$ be binary structures.

Reflexive: Let $\iota : S \to S$ be the identity map. Then ι maps S one to one onto S and for $a, b \in S$, we have $\iota(a * b) = a * b = \iota(a) * \iota(b)$, so ι is an isomorphism of S with itself, that is $S \simeq S$.

Symmetric: If $S \simeq S'$ and $\phi : S \to S'$ is an isomorphism, then by Exercise 26, $\phi^{-1} : S' \to S$ is an isomorphism, so $S' \simeq S$.

Transitive: Suppose that $S \simeq S'$ and $S' \simeq S''$, and that $\phi : S \to S'$ and $\psi : S' \to S''$ are isomorphisms. By Exercise 27, we know that $\psi \circ \phi : S \to S''$ is an isomorphism, so $S \simeq S''$.

29. Let $\langle S, * \rangle$ and $\langle S', *' \rangle$ be isomorphic binary structures and let $\phi : S \to S'$ be an isomorphism. Suppose that $*$ is commutative. Let $a', b' \in S'$ and let $a, b \in S$ be such that $\phi(a) = a'$ and $\phi(b) = b'$. Then $a' *' b' = \phi(a) *' \phi(b) = \phi(a * b) = \phi(b * a) = \phi(b) *' \phi(a) = b' * a'$, showing that $*'$ is commutative.

30. Let $\langle S, * \rangle$ and $\langle S', *' \rangle$ be isomorphic binary structures and let $\phi : S \to S'$ be an isomorphism. Suppose that $*$ is associative. Let $a', b', c' \in S'$ and let $a, b, c \in S$ be such that $\phi(a) = a', \phi(b) = b'$ and $\phi(c) = c'$. Then

$$\begin{aligned} (a' *' b') *' c' &= (\phi(a) *' \phi(b)) *' \phi(c) = \phi(a * b) *' \phi(c) = \phi((a * b) * c)) \\ &= \phi(a * (b * c)) = \phi(a) *' \phi(b * c) = \phi(a) *' (\phi(b) *' \phi(c)) = a' *' (b' *' c'), \end{aligned}$$

showing that $*'$ is associative.

31. Let $\langle S, * \rangle$ and $\langle S', *' \rangle$ be isomorphic binary structures and let $\phi : S \to S'$ be an isomorphism. Suppose that S has the property that for each $c \in S$ there exists $x \in S$ such that $x * x = c$. Let $c' \in S'$, and let $c \in S$ such that $\phi(c) = c'$. Find $x \in S$ such that $x * x = c$. Then $\phi(x * x) = \phi(c) = c'$, so $\phi(x) *' \phi(x) = c'$. If we denote $\phi(x)$ by x', then we see that $x' *' x' = c'$, so S' has the analagous property.

32. Let $\langle S, * \rangle$ and $\langle S', *' \rangle$ be isomorphic binary structures and let $\phi : S \to S'$ be an isomorphism. Suppose that S has the property that there exists $b \in S$ such that $b * b = b$. Let $b' = \phi(b)$. Then $b' *' b' = \phi(b) *' \phi(b) = \phi(b * b) = \phi(b) = b'$, so S' has the analogous property.

33. Let $\phi : \mathbb{C} \to H$ be defined by $\phi(a + bi) = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for $a, b \in \mathbb{R}$. Clearly ϕ is one to one and onto H.